Payroll Register System

Category B Practice - Part A, Problem 5

Problem summary

FullTime pay is a salary, PartTime pay is hours multiplied by rate, and Intern pay is a stipend. Store payslips in an array, then report total pay and the top earner.

Java solution

```
import java.util.*;
public class Main {
 static class Payslip {final String name,type;final double pay;Payslip(String n,String t,double p){name=n;type=t
  public String toString(){return "Payslip[name="+name+", type="+type+", pay="+fmt(pay)+"]";}}
 static String fmt(double n){return n==Math.rint(n)?Long.toString((long)n):String.format(Locale.US,"%.2f",n);}
 public static void main(String[] args){Scanner sc=new Scanner(System.in);List<Payslip> list=new ArrayList<>();
  while(sc.hasNextLine()){String line=sc.nextLine().trim();if(line.isEmpty())continue;String[] p=line.split("\\s
   switch(p[0]){case "FullTime":list.add(new Payslip(p[1],"FullTime",Double.parseDouble(p[2])));break;
    case "PartTime":list.add(new Payslip(p[1],"PartTime",Double.parseDouble(p[2])*Double.parseDouble(p[3])));bre
    case "Intern":list.add(new Payslip(p[1],"Intern",Double.parseDouble(p[2])));break;default:System.out.println
  }
  Payslip[] slips=list.toArray(new Payslip[0]);double total=0;Payslip top=null;
  for(Payslip s:slips){System.out.println(s);total+=s.pay;if(top==null||s.pay>top.pay)top=s;}
  System.out.println("Total "+fmt(total));if(top!=null)System.out.println("top earner "+top.name);
 }
}
```

Example input

```
FullTime Asha 50000
PartTime Ravi 80 300
Intern Neha 15000
```

Expected output

```
Payslip[name=Asha, type=FullTime, pay=50000]
Payslip[name=Ravi, type=PartTime, pay=24000]
Payslip[name=Neha, type=Intern, pay=15000]
Total 89000
top earner Asha
```

Complexity

Time O(n) for summing and finding the top earner. Space O(n) for the payslip collection and array.